MPhil Thesis: Methodology

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Contents

1 Methodology

1.1 Empirical Strategy

The empirical analysis takes on three stages. First, an event-study specification is utilised to estimate the aggregate innovation response to Medicaid exposure across therapeutic classes. PDL exposure is then incorporated to examine whether the innovation response differs depending on firms' exposure to Medicaid procurement mechanisms. Third, the post-expansion innovation response is allowed to vary for each therapeutic area in order to assess heterogeneity in innovation responses. The main outcomes are patent counts and FDA approvals, measured at the firm-therapeutic-class level.

The specifications are motivated by the market-size channel in pharmaceutical innovation, where an increase in expected demand should increase the expected return to R&D and therefore raise innovation incentives (Dubois et al., 2015; Finkelstein, 2004). Since the additional demand flows through a public buyer with rebate authority and formulary control, the Medicaid setting complicates this result. For this reason, the empirical analysis first aims to specifically examine whether the innovation response differs with a firm's exposure to the PDL.

1.2 Identification

The empirical strategy treats the ACA Medicaid expansion as a national shock to the expected profitability of drugs in therapeutic classes with different levels of pre-existing Medicaid exposure. While the expansion itself was adopted at the state level, innovation in the pharmaceutical industry is not a state-specific decision. Pharmaceutical firms develop drugs for the national markets, and as a result, the expected returns depend on anticipated sales across the United States as opposed to the realised expansion within any single expansion state. For this reason the empirical analysis does not leverage the staggered state adoption policy as the main source of variation. Instead, 2014 is treated as the relevant policy shock for innovation outcomes.

The expansion increased the number of individuals covered by Medicaid, and therefore increased the volume of prescriptions to be reimbursed through Medicaid. Since some therapeutic areas in Medicaid accounted for a larger share of prescriptions prior to the expansion, these areas were more exposed to the market-size shift compared with classes in which Medicaid played a smaller role. This indicates the shock did not hit all therapeutic classes equally. Therefore, the identifying variation stems from cross-class differences in pre-expansion Medicaid exposure. Firms operating in high-exposure classes are interpreted as facing a larger Medicaid demand shock after 2014. If the expansion increased expected demand in high-Medicaid classes, standard market-size models would predict a stronger innovation response in those classes after 2014 relative to less-exposed classes.

The main identifying assumption for each of the research questions is parallel trends. That is, absent the ACA Medicaid expansion, innovation outcomes in high and low Medicaid-exposure classes would have followed a similar trajectory over time. In this setting, this means that patenting and FDA approvals in high-exposure classes would have evolved similarly to low-exposure classes had the Medicaid expansion not occurred.

This is plausible since the constructed Medicaid exposure measure is largely determined by the patient population using drugs in each therapeutic class before 2014. Some classes have higher Medicaid utilisation due to treating conditions that are more prevalent amongst lower income Medicaid-eligible populations. This does not necessarily imply that those classes were on different innovation trajectories before the reform. Instead, innovation trends are likelier to depend on factors such as scientific opportunity. Consequently, there is no reason to expect classes with different exposure levels to Medicaid to have innovation outcome divergence absent the ACA expansion.

Since pharmaceutical R&D operates over long development cycles, patents and FDA approvals observed in a given year typically reflect investment decisions made several years prior. However, this could indicate that does make the identification timing more specific, as there could be a divergence in innovation trajectories prior to the expansion. The event-study specification assesses this by estimating pre-2014 coefficients - small and statistically insignificant coefficients support the parallel trends assumption further as it implies that classes with differing Medicaid exposures followed similar trajectories prior to the reform.

Identification also requires that there were no contemporaneous shocks around 2014 that differentially affected innovation in high-Medicaid-exposure classes. Year fixed effects are used to absorb shocks to the pharmaceutical industry such as FDA regulation changes or industry-wide shifts in R&D that affect all therapeutic areas within a given year. Given the fixed-effects, a confounding shock would therefore need to occur around 2014 and affect some therapeutic classes more than others. There are no clear candidates for such a shock, and other ACA provisions, including changes to private insurance markets and coverage mandates, affected the pharmaceutical sector more broadly and their effect would be absorbed by year fixed effects.

The specifications include firm-class fixed effects and year fixed effects. Firm-class fixed effects absorb time-invariant differences in innovation levels across firm-class pairs. This is important since firms may specialise in particular therapeutic areas - separate firm and class fixed effects may not be able to capture this source of heterogeneity. Year fixed effects absorb aggregate shocks to pharmaceutical innovation that are common across classes in a given year. Standard errors are clustered at the firm level to account for serial correlation in innovation outcomes within firms over time, since a firm's current innovation activity may influence its future innovation behaviour.

1.3 Construction of Key Variables

1.3.1 Ex-Ante Medicaid Market Share

Following the empirical framework established by Garthwaite et al. (2021) in their study of Medicaid expansions, the main exposure measure in this paper is based on ex-ante Medicaid market shares, calculated using prescription shares prior to the ACA Medicaid expansion. For each therapeutic class c, this predetermined variable is calculated as:

$$\text{MedicaidShare}_c = \frac{\sum_{t=2010}^{2013} \text{MedicaidPrescriptions}_{c,t}}{\sum_{t=2010}^{2013} \text{TotalPrescriptions}_{c,t}} \tag{1}$$

Classes with higher pre-expansion Medicaid shares are interpreted as those more exposed to the demand consequences of expansion and ex-ante construction avoids using post-treatment outcomes to define treatment intensity. Therapeutic classes may differ systematically in patient demographics or product characteristics, but these differences are likely to be persistent over time, thus the firm-class fixed-effects structure absorbs such time-invariant sources of heterogeneity.

1.3.2 Ex-Post Medicaid Market Share

As a complement to the ex-ante measure, an ex-post exposure proxy based on realised demand shifts in the early years of the reform is also calculated. For each therapeutic class, this is constructed as:

$$\text{MedicaidShare}_c = \frac{\sum_{t=2014}^{2015} \text{MedicaidPrescriptions}_{c,t}}{\sum_{t=2014}^{2015} \text{TotalPrescriptions}_{c,t}} \tag{2}$$

This captures the extent to which a therapeutic class actually experienced increased Medicaid demand in the immediate post-expansion period. Since this measure is closer to the realised response of the market, it may reflect the environment faced by firms in the Medicaid market after expansion. Given that this measure is not predetermined, estimates using this variable should be interpreted as a robustness check.

1.3.3 PDL Exposure

To examine exposure to the Medicaid pricing mechanisms, a measure of preferred drug list exposure for firm f in class c is constructed using the universe of drugs d:

$$\text{PDL\_Exposure}_{f,c} = \frac{\sum_d \mathbf{1}\{d \in f, c\} \cdot \mathbf{1}\{d \in \text{PDL}\}}{\sum_d \mathbf{1}\{d \in f, c\}} \tag{3}$$

Higher levels of PDL exposure indicate that a larger share of a firm's drug portfolio in a therapeutic class is subject to formulary selection and the pricing environment associated with preferred placement on the PDL. Empirically, this variable is used as a proxy for exposure to procurement institutions rather than as a direct measure of rebates or negotiated prices.

1.3.4 PDL Counterfactual Measures

Two time-invariant firm-class measures are constructed from the Texas Vendor Drug Program formulary data for use in the counterfactual analysis specification. Each is defined as the share of a firm's NDCs in class c satisfying a given PDL criterion, averaged over the baseline period 2010 to 2014:

$$\text{PT\_Pass}_{f,c} = \frac{\sum_d \mathbf{1}\{d \in f, c\} \cdot \mathbf{1}\{\text{PT\_Stage\_Pass}_d = 1\}}{\sum_d \mathbf{1}\{d \in f, c\}} \tag{4}$$

$$\text{Price\_Pass}_{f,c} = \frac{\sum_d \mathbf{1}\{d \in f, c\} \cdot \mathbf{1}\{\text{Price\_Stage\_Pass}_d = 1\}}{\sum_d \mathbf{1}\{d \in f, c\}} \tag{5}$$

These proportions are treated as time-invariant firm-class characteristics. Drugs with no match in the Texas formulary are assigned a value of zero for all three measures.

1.4 Research Questions

1.4.1 RQ1: Overall Innovation Response

The first research question asks whether therapeutic classes with greater Medicaid exposure experienced different innovation trajectories following the ACA Medicaid expansion. This is estimated using the following event-study specification:

$$Y_{f,c,t} = \alpha_{f,c} + \lambda_t + \sum_{\tau \neq 2013} \beta_\tau \left(\mathbf{1}\{t = \tau\} \times \text{MedicaidShare}_c \right) + \varepsilon_{f,c,t} \tag{6}$$

where $Y_{f,c,t}$ is an innovation outcome for firm f, therapeutic class c, and year t. The main outcomes are patent counts and FDA approvals. MedicaidShare_c measures pre-expansion Medicaid exposure in class c, $\alpha_{f,c}$ denotes firm-by-class fixed effects, and λ_t denotes year fixed effects. Since the coefficient for 2013 is omitted, each β_τ captures the differential innovation response associated with higher pre-expansion Medicaid exposure in year τ, relative to 2013.

This specification is estimated using both fixed-effects ordinary least squares (FEOLS) and fixed-effects Poisson (FE-Poisson) models. The linear FEOLS model provides a baseline and accommodates the event-study structure directly. By plotting the linear interaction coefficients, the FEOLS specification provides a straightforward, visually intuitive assessment of both the parallel trends assumption and the dynamic evolution of the treatment effect over time.

Innovation outcomes, such as patent grants and FDA approvals, are non-negative, and negatively skewed integer counts. Given this underlying data-generating process, the specification is also estimated using a FE Poisson model. The Poisson estimator is suitable in this setting since it can handle count data and zero-valued observations without requiring additional adjustments in functional form, and according to Wooldridge (1999) provided the conditional mean is correctly specified, the Poisson estimator provides consistent estimates even if the underlying data exhibits significant overdispersion. Therefore, estimating both linear and Poisson specifications allows the analysis to assess whether the main findings are sensitive to the treatment of the count nature of the outcomes.

Both ex-ante and ex-post exposure measures are used to measure MedicaidShare_c. The ex-ante measure is based on pre-2014 Medicaid prescription shares and captures exposure that is predetermined with respect to the policy change. The ex-post measure instead utilises realised post-expansion demand shifts and is interpreted more cautiously, as it potentially contains the contemporaneous market response induced by the ACA.

1.4.2 RQ2: Market Size and PDL Exposure

The second research question asks whether the innovation response to Medicaid expansion differs depending on a firm's exposure to the PDL institutions. The event-study specification takes on form:

$$
\begin{aligned}
Y_{f,c,t} = \alpha_{f,c} + \lambda_t \\
&+ \sum_{\tau \neq 2013} \beta_\tau \left(\mathbf{1}\{t=\tau\} \times \text{MedicaidShare}_c\right) \\
&+ \sum_{\tau \neq 2013} \gamma_\tau \left(\mathbf{1}\{t=\tau\} \times \text{PDL\_Exposure}_{f,c}\right) + \varepsilon_{f,c,t}
\end{aligned}
\tag{7}
$$

where $\text{PDL\_Exposure}_{f,c}$ measures the extent to which firm f's portfolio in class c is exposed to Medicaid's formulary institutions. This is constructed as the share of the firm's drugs in a given therapeutic class that appear on the PDL. The coefficients on the MedicaidShare_c interactions capture how innovation varies with Medicaid demand exposure in each year relative to 2013. The coefficients on the $\text{PDL\_Exposure}_{f,c}$ interactions capture how innovation varies with firms' exposure to Medicaid formulary institutions over the same period.

While this model does not provide a clean structural decomposition of market size and pricing pressures, PDL exposure is utilised as a proxy rather than a direct measure of rebate extraction or bargaining intensity. Higher PDL exposure being associated with weaker post-expansion innovation responses would be consistent with the institutional setting of Medicaid weakening the standard market-size effect.

1.4.3 RQ3: Therapeutic-Class Heterogeneity

The third research question investigates whether the effect of the Medicaid expansion differs across therapeutic classes. This is important because Medicaid exposure, market structure, and PDL exposure may vary across therapeutic areas in ways that are not captured by the aggregate specification.

To examine this, a triple-interaction specification is specified. This allows the post-expansion effect of Medicaid exposure to vary by the ATC Level 1 class:

$$
Y_{f,c,t} = \alpha_f + \delta_c + \lambda_t + \sum_{a \in \mathcal{A}} \theta_a \left(\mathbf{1}\{t \geq 2014\} \times \text{MedicaidShare}_c \times \mathbf{1}\{c=a\}\right) + \varepsilon_{f,c,t} \tag{8}
$$

where a is a specific ATC within the universe of ATC Level 1 therapeutic areas $\mathcal{A}$, and θ_a captures the post-2014 differential response associated with Medicaid exposure in therapeutic area a. These estimates provide evidence on the heterogeneity of the response to the ACA and help assess whether the aggregate results are driven by particular therapeutic areas.

1.4.4 PDL Counterfactual Analysis

In order to examine the innovation mechanism more directly, a counterfactual design is estimated by leveraging the two-stage structure of Medicaid Preferred Drug List (PDL) placement. A drug receives preferred status if and only if it clears both a clinical (P&T) stage, and a pricing stage. Firms whose products pass only one stage provide a useful counterfactual group for examining whether innovation responses differ against firms that either pass both or none of the margins. This is tested for using the following triple-difference specification, which isolates the post-expansion innovation responses associated with passing each stage in from the response associated with securing full PDL placement:

$$
\begin{aligned}
Y_{f,c,t} = \alpha_{f,c} + \lambda_t &+ \beta_1 \left(\text{PT\_Pass}_{f,c} \times \text{Post}_t\right) \\
&+ \beta_2 \left(\text{Price\_Pass}_{f,c} \times \text{Post}_t\right) \\
&+ \beta_3 \left(\text{PT\_Pass}_{f,c} \times \text{Price\_Pass}_{f,c} \times \text{Post}_t\right) \\
&+ \varepsilon_{f,c,t}
\end{aligned}
\tag{9}
$$

where $\text{PT\_Pass}_{f,c}$, $\text{Price\_Pass}_{f,c}$ are proportions measuring the share of a firm's drugs that pass the clinical stage and the pricing stage for the PDL. The triple interacted term provides the proportion of a firm's drugs within a class that pass both stages jointly.

The coefficient β_1 captures the post-expansion change in innovation associated with a higher share of a firm's drugs passing the clinical stage but not securing preferred placement. β_2 measures the post-expansion change in innovation associated with a higher share of a firm's drugs passing the pricing stage but not meeting the clinical criteria for preferred placement. β_3 captures the additional innovation response associated with a higher share of drugs passing both stages and therefore securing preferred status.

This exercise is not intended to identify the full structural effect of the PDL process. However, it tests whether the innovation response differs across firms depending on their position within the PDL. If firms which pass both stages more often experience more favourable innovation responses, then this would suggest that preferred placement on the PDL affects the profitability of innovation.

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